



ACADEMIC DETAILS

PROGRAM	INSTITUTION	CGPA/%	YEAR
IDDD ¹ : Biological Sciences (BS) + Quantitative Finance (M. Tech.)	Indian Institute of Technology, Madras	8.75	2026
Class XII (CBSE)	Mount Litera Zee School, Unnao	93.8%	2020
Class X (ICSE)	Dr. Virendra Swarup Education Centre, Kanpur	95%	2018
Scholarship	Awarded academic scholarship by the PowerGrid Corporation of India Limited , in the freshman year		

COURSEWORK

Principles of Economics	Corporate Finance	Economics of Industrial Organizations
Production Economics*	Functions of Several Variables	Probability and Statistics
Mathematical Finance*	Introduction to Programming	Supervised Machine Learning

PROFESSIONAL EXPERIENCE

Indian Institute of Technology, Kanpur Undergraduate Research Intern, Decision Lab May'23 – Jul'23	Exploring the impact of Trait Anxiety on Attention and Performance Guide: Prof. Arjun Ramakrishnan - Investigated the impact of tDCS² on attention span of trait anxious individuals through ERP³ (P300). - Designed & implemented a custom Flanker Task for the study and collected pre/post-tDCS EEG⁴ data. - Demonstrated an improvement of 5% to 20% in accuracy after only a single 30 minutes stim session. - Simulated the effect of multiple electric fields on the human brain to minimize brain tissue damage.
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PROJECTS

Artificial Immune System Biotech Club, Centre for Innovation Jun '23 – Apr '24	- Developed PSA⁵ models to detect Parkinson's disease using audio, handwriting and time series data - Pre-processed data using standardization and PCA; implemented PSA with Minkowski distance metric - Achieved accuracy of 93.63% for audio, 77.42% for handwriting data, and 73.08% for time series data - Obtained ROC score of 0.7796 for audio data and demonstrated the discriminative ability of the model
Crypto Trade Jan '24	- Developed a Bitcoin price prediction model using an LSTM neural network, analyzing 1m OHLC data - Trained model on 3 years of historical data, achieving a low test MAE loss of 0.2979 and no anomalies - Generated buy and sell trading signals based on technical indicators like RSI and the Bollinger Bands
Protein Unfolding Prof. Athi N. Naganathan Feb '23 – Mar '23	- Developed Python simulation employing Monte Carlo algorithm to analyse polymer unfolding paths - Implemented Metropolis criterion; calculated parameters: radius of gyration & end-to-end distance - Discovered decreasing interaction energy from -0.25 to -2.50 increased mean unfolding time by 256%
Portfolio Optimization Prof. Madhumathi R. Apr '24 – May '24	- Optimized Nifty Commodities Index portfolio, reducing risk by 20.2% while maintaining the returns - Constructed minimum variance and Markowitz portfolios and improved the Sharpe Ratio by 75.2% - Created insightful visualizations of the Efficient Frontier and Indifference Curve using plotly library
OFSS Financial Evaluation Prof. Arun Kumar G. Feb '24 – Apr '24	- Analysed Oracle Financial Services Software Ltd. financials via profitability, liquidity, efficiency ratios - Identified a slight decline in profitability and liquidity, but with an improved asset utilization efficiency - Highlighted the company's favourable dividend policy with payouts consistently above 90% of the PAT

VIRTUAL EXPERIENCES⁶

J.P. Morgan Chase and Co. Quantitative Research Jun '24	- Developed pricing model to estimate natural gas prices by using linear regression on historical data - Developed a python function to price storage contracts, factoring in dates, prices, rates, volume, costs - Implemented quantization algorithm using dynamic programming to map FICO⁷ scores to the ratings - Built Random Forest model to predict loan default and expected loss with an accuracy of over 90% - Optimized credit rating buckets by maximizing log-likelihood function, improving default estimates
HSBC Global Banking and Markets Jun '24	- Analysed potential M&A targets for expansion into Asia; identified top candidate fulfilling our criteria - Conducted credit analysis on \$400M acquisition, prepared a Debt Capital Market update using Excel - Ensured that the deal meets HSBC's internal thresholds of <5x leverage and >2x interest coverage - Developed Excel model to analyse market data & recommend a hedge against stock market declines

POSITIONS OF RESPONSIBILITY

Coordinator Biotech Club Centre for Innovation Jun'23 – May'24	- Successfully conducted a demonstrative session during the Summer School 2023, conducted by CFI - Session on Transcranial Electrical Stimulation was attended by 200+ students from across the world - As a founding member of Biotech Club, solely responsible for handling public relations & outreach - Responsible for engaging over 1000+ students and ensuring smooth logistics of various club events
Manager Career Development Cell Jun'22 – May'23	- Curated the Higher Studies Bluebook for the students interested in pursuing research in their fields - Conducted extensive research into graduate programs & curricula of major international universities - Facilitated smooth conduction of the placement preparation sessions conducted by Cantilever Labs
SAATHI Mentor	Appointed as a student mentor to 3 freshmen and ensured that they acclimatise to the college life

EXTRACURRICULAR ACTIVITIES

National Cadet Corps	Served as a cadet at the 4 Tamil Nadu Air (Tech) squadron of NCC, Air Force in the Chennai B group
Cocurricular	Awarded GESE ⁸ certificate in ESOL ⁹ International (Speaking and Listening) of the Common European Framework of Reference for Languages with Distinction , conducted by Trinity College, London .

1: IDDD: Interdisciplinary Dual Degree; 2: tDCS: Transcranial Direct Current Stimulation; 3: ERP: Event Related Potentials; 4: EEG: Electroencephalograph; 5: PSA: Positive Selection Algorithm; 6: Forage Work Simulation; 7: FICO: Fair Isaac Corporation; 8: GESE: Graded Examination in Spoken English; 9: ESOL: English for Speakers of Other Languages; *: Ongoing Course